

Zen Nguyen

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TECHNICAL SKILLS

- **Languages:** C++, C, Go, Python, Java, TypeScript, JavaScript
- **Backend & Systems:** gRPC, RESTful APIs, Raft Consensus Algorithm
- **Infrastructure & DevOps:** Docker, AWS, Linux
- **Testing & Validation:** Software Unit Test Development, Software Verification and Analysis, Software-level Test Case Development, Software-level Test Script Development, Automated Software Verification Test Procedures

TECHNICAL PROJECTS

TitanStore (Distributed Raft Database)

Jan 2026 - Present

Personal Project · Go, Raft Consensus Algorithm, TCP Sockets, WAL

- Ensured data durability and enabled node recovery after failures without data loss by engineering a crash-safe Write-Ahead Log (WAL) with atomic snapshots.
- Developed a distributed key-value database in Go from scratch, implementing the **Raft Consensus Algorithm** for leader election and fault-tolerant log replication across a cluster of nodes.
- Designed and implemented a high-throughput raw TCP server on port 6001, parsing a custom text protocol supporting SET, GET, and ERR NOT_LEADER operations with non-blocking I/O.
- Achieved fault tolerance and high availability through the implementation of the **Raft Consensus Algorithm**, ensuring reliable operation even with node failures.

SFU Course Tracker (sfucourseplanner.me)

Nov 2025 - Jan 2026

Personal Project · Python, Docker, AWS, FastAPI, SQLAlchemy

- Migrated full-stack platform from Azure to AWS (EC2), implementing **Docker** container orchestration to optimize deployments and reduce infrastructure costs.
- Engineered a scraping pipeline using Asyncio and HTTPX with semaphore-based rate limiting, increasing data throughput by 10x while respecting server constraints.
- Designed a normalized schema using SQLAlchemy with JSON-type columns to store recursive tree structures, optimizing read performance for complex queries.
- Developed a custom parser to tokenize and evaluate nested boolean prerequisite strings, converting unstructured text into a deterministic Abstract Syntax Tree (AST).

TitanSwarm (Autonomous AI Co-Pilot)

Jan 2026 - Present

Personal Project · Python, LangChain, FAISS, Streamlit, Gemini API

- Built a zero-hallucination RAG (Retrieval-Augmented Generation) engine using LangChain, Gemini APIs, and a sandboxed FAISS Vector Store, delivering uniquely tailored, ATS-optimized PDF resumes for each job using only verified user data.
- Architected an autonomous AI Co-Pilot in Python that automates end-to-end job application workflows, enabling the discovery, extraction, and parsing of real-time Software Engineering postings via concurrent JobSpy scrapers across LinkedIn and Indeed.
- Applied **Test-Driven Development (TDD)** with a 113-test suite using Pytest, mocking, and async fixtures to validate all components including the scraper, AI tailor, PDF generator, and repository layer.

EDUCATION

Bachelor of Science, Computing Science

May 2025 - Present

Simon Fraser University

- CGPA: 3.74 / 4.33
- Relevant Coursework: Discrete Math (A+), Computer Systems(A), Data Structures & Programming (B+), Linear Algebra(A-).
- Langara College | Associate of Science, Computer Science Jan 2024 - Apr 2025